# Classical Mechanics

# States of a system

- State
  - Minimum number of variables required to fully determine the future trajectory of a system

$$q = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_n \end{bmatrix}$$

- Non-Holonomic
  - The elements in the derivative $\dot{q}$ are coupled by constrains
  - *Some path cannot be taken even if all the states along the trajectory are permissible
  - E.g. Unicycle
    - $\text{atan}(\dot{y}, \dot{x}) = \theta$
- Degrees of Freedom
  - The difference between number of states and the non-holonomic constrains

# Hamilton's principle

- The Lagrangian of the real trajectory of a system is a local optimum

$$\delta J = \delta \left[ \int_{t_1}^{t_2} L(x(t), \dot{x}(t), t)\, dt \right] = 0$$

- Euler-Lagrange equation

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}_i} - \frac{\partial L}{\partial x_i} = 0 \quad \forall i$$

# Proof: Euler-Lagrange

$$\delta\left[\int_{t_1}^{t_2} L(x(t),\dot{x}(t),t)\,dt\right]$$

$$= \int_{t_1}^{t_2} \sum_i \left(\frac{\partial L}{\partial x_i}\delta x_i + \frac{\partial L}{\partial \dot{x}_i}\delta \dot{x}_i\right) dt$$

$$= \sum_i \left(\int_{t_1}^{t_2} \frac{\partial L}{\partial x_i}\delta x_i\,dt + \int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{x}_i}\delta \dot{x}_i dt\right)$$

$$= \int_{t_1}^{t_2} \sum_i \left(\frac{\partial L}{\partial x_i}\delta x_i - \frac{d}{dt}\frac{\partial L}{\partial \dot{x}_i}\delta x_i\right) dt$$

$$= \int_{t_1}^{t_2} \sum_i \left(\frac{\partial L}{\partial x_i} - \frac{d}{dt}\frac{\partial L}{\partial \dot{x}_i}\right)\delta x_i dt \;= 0$$

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{x}_i}\delta \dot{x}_i dt = \int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{x}_i}\frac{d}{dt}\delta x_i dt$$

$$= \left|\frac{\partial L}{\partial \dot{x}_i}\delta x_i\right|_{t_1}^{t_2} \quad \text{By part}$$

$$= -\int_{t_1}^{t_2} \frac{d}{dt}\frac{\partial L}{\partial \dot{x}_i}\delta x_i dt \quad \text{No variation at boundary}$$

For this to be true for all possible variations $\delta x_i$

$$\frac{\partial L}{\partial x_i} - \frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = 0 \quad \forall i$$

# Example : Mechanical System of Particles

$$T \coloneqq \sum_k \frac{1}{2} m_k (x^2 + y^2 + z^2) \qquad V \coloneqq V(x_1, y_1, z_1 \cdots, x_k, y_k, z_k, \cdots, t)$$

$$L = T - V$$

$$\frac{d}{dt} m_k \dot{x}_k + \frac{\partial V}{\partial x_k} = 0 \qquad F_{x_k} + \frac{\partial V}{\partial x_k} = 0$$

$$\frac{d}{dt} m_k \dot{y}_k + \frac{\partial V}{\partial y_k} = 0 \qquad F_{y_k} + \frac{\partial V}{\partial y_k} = 0$$

$$\frac{d}{dt} m_k \dot{z}_k + \frac{\partial V}{\partial z_k} = 0 \qquad F_{z_k} + \frac{\partial V}{\partial z_k} = 0$$

# Hamiltonian Function

Pontryagin H function in optimal control

$$\dot{x} = f(x, u)$$

Generalized momenta -> costate

$$H = \lambda^T f(x, u) - L$$

Costate dynamics

$$\dot{x}_i = \frac{\partial H}{\partial \lambda_i}, \qquad \dot{\lambda}_i = -\frac{\partial H}{\partial x_i}$$

- Generalized momenta

$$p_i \coloneqq \frac{\partial L}{\partial \dot{x}_i}$$

- Hamiltonian function

$$H \coloneqq \left(\sum_i p_i \dot{x}_i\right) - L$$

$$\frac{\partial H}{\partial p_i} = \dot{x}_i \qquad \frac{\partial H}{\partial x_i} = -\frac{\partial L}{\partial x_i} = -\dot{p}_i$$

Euler-Lagrange

$$\frac{\partial L}{\partial x_i} - \frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = 0$$

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} = \frac{\partial L}{\partial x_i}$$

$$\dot{p}_i = \frac{\partial L}{\partial x_i}$$

$$H = V(x, u, t) + \lambda^T f(x, u)$$

- Canonical Form

$$\dot{x}_i = \frac{\partial H}{\partial p_i}, \dot{p} = -\frac{\partial H}{\partial x_i}$$

# Poison Brackets

- Dynamics of arbitrary quantity as a function of $x, p$ (but not $t$)

$$F(x,p)$$

$$\frac{dF}{dt} = \sum_i \left( \frac{\partial F}{\partial x_i}\frac{dx_i}{dt} + \frac{\partial F}{\partial p_i}\frac{dp_i}{dt} \right)$$

$$= \sum_i \left( \frac{\partial F}{\partial x_i}\frac{\partial H}{\partial p_i} - \frac{\partial F}{\partial p_i}\frac{\partial H}{\partial x_i} \right) \coloneqq \{F, H\}$$

Poison Brackets

# Energy

- If H is not an explicit function of t
  - Let $F = H$,

$$\frac{dH}{dt} = \{H, H\} = \sum_i \left( \frac{\partial H}{\partial x_i} \frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i} \frac{\partial H}{\partial x_i} \right) = 0$$

$$H(x, p) = E$$

# Example: Mechanical Energy

$$L = T - V \qquad T \coloneqq \frac{1}{2}\sum_k m_k(x^2 + y^2 + z^2)$$

$$\begin{aligned}
H &= p_x\dot{x} + p_y\dot{y} + p_z\dot{z} - L \\
&= p_x\dot{x} + p_y\dot{y} + p_z\dot{z} - T + V \\
&= \frac{\partial L}{\partial \dot{x}}\dot{x} + \frac{\partial L}{\partial \dot{y}}\dot{y} + \frac{\partial L}{\partial \dot{z}}\dot{z} - T + V \\
&= \frac{\partial T}{\partial \dot{x}}\dot{x} + \frac{\partial T}{\partial \dot{y}}\dot{y} + \frac{\partial T}{\partial \dot{z}}\dot{z} - T + V \\
&= mx^2 + my^2 + m^2 - T + V \\
&= 2T - T + V \\
&= T + V = E
\end{aligned}$$

# Linear and angular momentum